

Unit IV Question Bank

MCQs

1. In a standard sequential neural network, how does each layer process data?
 - A. It processes all data in parallel
 - B. It receives the previous layer's output and passes the result to the next
 - C. It skips layers to speed up training
 - D. It processes data independently of other layers

Correct Answer: B

2. What is a primary limitation of very deep sequential convolutional networks?
 - A. They always have too few parameters
 - B. They cannot be used for image classification
 - C. Performance decreases as further layers are added beyond a certain point
 - D. They only work with linear activation functions

Correct Answer: C

3. The "shattered gradients" phenomenon refers to which situation in deep networks?
 - A. Gradients becoming too large to compute
 - B. Gradients becoming exactly zero at every layer
 - C. Tiny changes in input resulting in completely different, uncorrelated gradients
 - D. Gradients only moving in one direction during optimization

Correct Answer: C

4. What is the core idea behind a "residual block"?
 - A. Removing layers to simplify the model
 - B. Adding the input of a layer back to its processed output
 - C. Using only linear transformations
 - D. Multiplying the input by the output

Correct Answer: B

5. In the residual network equation $h_1 = x + f_1[x, \varphi_1]$, what does x represent?
 - A. The residual connection or input
 - B. The learned weight matrix
 - C. The activation function
 - D. The output of the entire network

Correct Answer: A

6. Which of the following is true regarding the size of inputs and outputs in a residual function f_k ?
 - A. The output must be twice the size of the input
 - B. The output must be smaller than the input
 - C. The output must be the same size as the input to allow addition
 - D. The size does not matter in residual blocks

Correct Answer: C

7. How can a residual network be interpreted in terms of "paths"?
 - A. It creates a single, very long path
 - B. It creates multiple paths of different lengths from input to output
 - C. It eliminates all paths except the shortest one
 - D. It uses circular paths to reuse data

Correct Answer: B

8. Why do residual links suffer less from shattered gradients?
- A. Because they remove all non-linearities
 - B. Because short paths allow gradients to contribute directly to the output
 - C. Because they use fewer parameters than sequential networks
 - D. Because they only use 1 x 1 convolutions

Correct Answer: B

9. What happens if a residual block lacks a nonlinear activation function like ReLU?
- A. It becomes more accurate
 - B. The entire network becomes a linear transformation
 - C. The gradients will always explode
 - D. It cannot be trained using SGD

Correct Answer: B

10. Why is it typical to apply the activation function *before* the linear transformation in modern residual blocks?
- A. To make the network slower
 - B. To ensure the output is always negative
 - C. To prevent each block from only increasing the input values
 - D. To reduce the number of channels

Correct Answer: C

11. Without careful initialization, what can happen to intermediate values in a deep network?
- A. They always stay at exactly one
 - B. They can increase or decrease exponentially
 - C. They become independent of the input
 - D. They automatically normalize themselves

Correct Answer: B

12. Which initialization method is specifically designed for ReLU activations?
- A. Xavier initialization
 - B. He initialization
 - C. Zero initialization for all weights
 - D. Random initialization with mean 1

Correct Answer: B

13. In a residual network using He initialization, why does variance increase through the network?
- A. Because of the softmax layer
 - B. Because the variability of each added branch is uncorrelated, doubling the variance
 - C. Because the learning rate is too high
 - D. Because layers are being removed

Correct Answer: B

14. What is a common method to stabilize the forward and backward passes in residual networks?
- A. Using only sigmoid activations
 - B. Removing skip connections
 - C. Batch normalization

D. Increasing the dataset size

Correct Answer: C

15. What does Batch Normalization (BatchNorm) do to each activation?

- A. Multiplies it by a constant zero
- B. Shifts and rescales it based on the mean and variance of a batch
- C. Clips all values to be between 0 and 1
- D. Replaces the activation with its index

Correct Answer: B

16. During Batch Normalization, what are γ and δ ?

- A. Constants that never change
- B. Hyperparameters set before training
- C. Learned scales and offsets
- D. The input and output dimensions

Correct Answer: C

17. How are BatchNorm statistics handled at test time?

- A. They are recalculated for every single test image
- B. They are calculated across the training dataset and frozen
- C. They are set to zero and one respectively
- D. They are ignored entirely

Correct Answer: B

18. Which benefit does Batch Normalization provide regarding learning rates?

- A. It requires very small learning rates
- B. It allows for higher learning rates by smoothing the loss surface
- C. It makes the learning rate irrelevant
- D. It only works with a learning rate of 1.0

Correct Answer: B

19. How does Batch Normalization act as a form of regularization?

- A. It injects noise because normalization depends on other members of the batch
- B. It removes the need for a training set
- C. It strictly limits the weights to be small
- D. It forces the network to be shallow

Correct Answer: A

20. In a standard ResNet block, what is the typical sequence of operations?

- A. Conv -> ReLU -> BN
- B. BN -> ReLU -> Conv
- C. Softmax -> Conv -> BN
- D. Linear -> Dropout -> Conv

Correct Answer: B

21. What is the purpose of a "bottleneck" residual block?

- A. To make the network slower for better accuracy
- B. To make more efficient use of parameters
- C. To stop the flow of data at certain layers
- D. To increase the spatial resolution of images

Correct Answer: B

22. A bottleneck residual block uses a 1 x 1 convolution primarily to:

- A. Increase the kernel size

- B. Reduce the number of channels
- C. Blur the image
- D. Replace the ReLU function

Correct Answer: B

23. The ResNet-200 model was specifically evaluated on which database?

- A. MNIST
- B. ImageNet
- C. CIFAR-10
- D. COCO

Correct Answer: B

24. ResNet-200 was notable for being one of the first networks to:

- A. Use only 10 layers
- B. Exceed human performance on ImageNet
- C. Use no activation functions
- D. Be trained on a single CPU

Correct Answer: B

25. What is the current state-of-the-art model type for image classification?

- A. Standard CNNs
- B. VGG networks
- C. Transformers
- D. AlexNet

Correct Answer: C

26. How does a DenseNet differ from a ResNet in its connection style?

- A. It adds the input to the output
- B. It concatenates the input and the output representations
- C. It subtracts the input from the output
- D. It has no skip connections

Correct Answer: B

27. What happens to the number of channels in a DenseNet as data moves through a dense block?

- A. It decreases exponentially
- B. It increases as outputs from previous layers are concatenated
- C. It stays exactly the same
- D. It fluctuates randomly

Correct Answer: B

28. In DenseNet, why is the chain of concatenation broken periodically?

- A. Because representations have different sizes after downsampling
- B. To make the network harder to train
- C. Because the computer runs out of memory immediately
- D. To prevent the use of 1 x 1 convolutions

Correct Answer: A

29. What is a "U-Net" primarily designed for?

- A. Stock market prediction
- B. Medical image segmentation
- C. Playing board games
- D. Audio compression

Correct Answer: B

30. What is the structural characteristic of a U-Net?

- A. It is a single long line of layers
- B. An encoder-decoder architecture with concatenations between them
- C. It only has one layer
- D. It uses only fully connected layers

Correct Answer: B

31. In the original U-Net, why was cropping necessary before concatenation?

- A. To save disk space
- B. Because "valid" convolutions made the upsampled version smaller than the encoder version
- C. To remove background noise
- D. To change the color of the image

Correct Answer: B

32. How does an Hourglass network differ from a U-Net in its skip connections?

- A. It uses no skip connections
- B. It applies further convolutional layers and adds the result instead of concatenating
- C. It skips every other layer
- D. It only uses max pooling in skip connections

Correct Answer: B

33. What is a common application for Stacked Hourglass networks?

- A. Text translation
- B. Human pose estimation
- C. Weather forecasting
- D. Sentiment analysis

Correct Answer: B

34. In pose estimation, what does the network predict for each joint?

- A. A single (x, y) coordinate pair
- B. A "heatmap" where the peak indicates the joint position
- C. The color of the joint
- D. The name of the joint

Correct Answer: B

35. Up to how many layers can a ResNet architecture be extended and still train effectively?

- A. 50 layers
- B. 100 layers
- C. 1000 layers
- D. 10 layers

Correct Answer: C

36. Which finding suggests that depth is not the *only* factor in ResNet performance?

- A. Shallower, wider residual networks can outperform deeper, narrower ones
- B. Deeper networks are always faster to train
- C. Residual networks cannot be used with few layers
- D. Only narrow networks work for image classification

Correct Answer: A

37. What is one view of how a very deep residual network actually functions?

- A. As a single massive matrix
- B. As a combination or ensemble of shallower networks
- C. As a sequence of random numbers
- D. As a linear regression model

Correct Answer: B

38. How does the loss surface of a residual network compare to a similar network without skip connections?
- A. It is much more jagged
 - B. It is smoother and more predictable
 - C. It has more local minima
 - D. It is exactly the same

Correct Answer: B

39. What is the benefit of a smoother loss surface?
- A. It makes training impossible
 - B. It facilitates learning and makes performance more robust to parameter errors
 - C. It prevents the use of GPUs
 - D. It requires more training data

Correct Answer: B

40. What is the purpose of adding a small number ϵ in the BatchNorm formula?
- A. To increase accuracy
 - B. To prevent division by zero
 - C. To make the mean zero
 - D. To slow down the calculation

Correct Answer: B

41. In a convolutional network, BatchNorm statistics are computed over:
- A. Only the first image in the batch
 - B. Both the batch and the spatial positions
 - C. Only the center pixel
 - D. Only the weights

Correct Answer: B

42. Why is Batch Normalization considered slightly "inefficient"?
- A. It creates redundancy in weight parameters and then adds extra parameters to compensate
 - B. It removes all parameters from the network
 - C. It requires two weights for every single pixel
 - D. It only works on CPUs

Correct Answer: A

43. What happens to effective depth with BatchNorm at the start of training?
- A. It is effectively less deep because later layers are close to the identity
 - B. It is immediately 1000 layers deep
 - C. It starts at zero depth
 - D. It depends on the image size

Correct Answer: A

44. How does the variance grow in a residual network *with* Batch Normalization at initialization?
- A. Exponentially

- B. Linearly
- C. It doesn't grow at all
- D. It decreases

Correct Answer: B

45. What sequence of layers typically starts a ResNet?
- A. A residual block
 - B. A standard linear (or convolutional) transformation
 - C. A ReLU layer alone
 - D. A MaxPool layer

Correct Answer: B

46. DenseNet's ability to reuse processing from earlier layers makes it:
- A. Use more parameters than ResNet
 - B. Perform better for a comparable parameter count
 - C. Impossible to train on small datasets
 - D. Only useful for text data

Correct Answer: B

47. In the U-Net, what operation increases spatial resolution in the decoder?
- A. MaxPool
 - B. Transposed Convolution (TConv)
 - C. Standard Convolution
 - D. Identity mapping

Correct Answer: B

48. What is the main finding regarding "short paths"?
- A. They are useless for training
 - B. Gradients through them are better behaved
 - C. They are the only paths that exist
 - D. They always cause gradients to explode

Correct Answer: B

Descriptive Questions

1. Explain the "Shattered Gradients" phenomenon and its impact on training deep sequential networks.
2. Describe the architecture and interpretation of a Residual Block. How does it differ from a standard sequential layer?
3. Analyze why standard initialization (like He initialization) is insufficient for very deep residual networks and how Batch Normalization addresses this.
4. Compare and contrast the ResNet and DenseNet architectures in terms of how they handle information flow.
5. Explain the "Unraveled Network" perspective of Residual Networks. How does this explain their superior performance?